

Research Statement

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1. Overview

My research focuses on the study of epistemic systems, or “system-oriented social epistemic”, which lies at the intersection of *social epistemology*, *philosophy of science*, and *philosophy of AI*. The overarching theme of my research is the interplay among **individual rationality**, **technology**, and **the institutional organization of collective inquiry**.

Epistemic activities of our time are inevitably shaped by technologies and institutions. This reality urges epistemology to go beyond the traditional individualistic perspective. My work draws on formal tools to examine individual and collective inquiries against the technological and institutional arrangements that shape epistemic activities. In my Ph.D. dissertation, I analyzed three different epistemic systems: expert system, voting, and financial markets, from a systematic perspective while emphasizing the interplay between individuals and institutions.

I continue to develop this line of work in relation to epistemic aggregation and expert identification (Section 2), while extending it to recent developments in AI. My current research examines both the social epistemic consequences of AI (Section 3) and what AI and social epistemology can learn from one another (Section 4). I also have a longer-term project that examines the rational and pragmatic comparison of conceptual frameworks in science and in social life (Section 5).

2. Individual Rationality and Epistemic Institutions

2.1. Epistemic Aggregation

I’m interested in how individual rationality and collective rationality interact in the context of epistemic aggregation, where individual judgments are aggregated into a collective judgment. In a recent paper¹, I address this question in respond to the dispute over sincere and strategic voting. I showed that rational voting models in economics rely on equilibrium concepts to justify the rationality of strategic voting. Such justifications implicitly assume unrealistic epistemic conditions about individuals. In rejecting these assumptions, I defend sincere voting on rational grounds. The paper is a substantive defense of sincere voting regarding individual rationality in epistemic aggregation but also raises an important methodological question about game theory. Namely, under what conditions can certain solution concepts of game theory ground claims about individual and collective rationality.

¹ The paper is currently under review at Philosophical Studies, with the title “The Inapplicability of Nash Equilibrium to the Condorcet Jury Theorem or: Why I’d rather vote sincerely and so should you”. Draft is available here.

My current work extends the study of epistemic aggregation into the context of AI-mediated group inquiries. See Section 3.1.

2.2. Expert Identification

Most, if not all, people living in a modern society obtain most of her knowledge from others, say textbooks, documentary films, newspapers, school teachers, etc. Finding the correct source to trust, known as the problem of expert identification, is crucial for everyone living in the modern theory. The problem of expert identification is traditionally treated as a paradigmatic problem of individual agents facing social evidence. In a paper², I argued that systematic perspective has much to say about the problem even in a one-expert-one-novice scenario. The seemingly individualistic decision of trusting experts is implicitly influenced by experts' strategic behaviors, which can and can only be counterbalanced by institutional arrangements. The general message is that all individuals must make their decisions against a background of epistemic infrastructure that one often fails to notice. The distinctive role of system-oriented social epistemology is to ensure that these background structures do not escape scrutiny. My current work extends the study of normative epistemic institution design into the era of AI. See Section 3.2.

3. Social Epistemic Consequences of AI

3.1. Algorithmic Monoculture

Algorithmic monoculture is the phenomenon in which many agents rely on the same, or highly similar, algorithms to form beliefs and decisions. Algorithmic monoculture can lead to epistemic homogeneity and harm collective inquiries and decision-making. The widespread adoption of a small number of large language model (LLM) families makes worries around this phenomenon especially pressing. Even users of different products may in fact rely on systems with overlapping data, retrieval sources, or model backbones.

My paper on algorithmic monoculture identifies one potential harm in the context of epistemic aggregation: an epistemic Braess paradox in which LLMs make individuals better informed while making a group less accurate. The formal analysis separates a gain in individual competence from a loss of independence as errors become correlated and examines when the latter outweighs the former. In assessing potential remedies, I identified the problem of an illusion of independence, where personalized LLM interactions can conceal overlap among agents' sources. This creates a new provenance problem for existing proposals of remedy and distinguishes LLMs from other shared sources of evidence.

The more general question is when and why epistemic diversity benefits collective inquiry, and when convergence is instead epistemically desirable. I plan to generalize the analysis of

² "Regulating Strategic Track-Record Building: Institutional Remedies for Individual Expert Identification", *Synthese*, 2026, volume 208, article 11.

epistemic diversity beyond that of evidence and experience, to include different methods and conceptual frameworks. How does widespread reliance on AI and different algorithmic culture (mono vs poly) change these forms of diversity and their epistemic value?

3.2. Epistemic Institutional Design under AI Disruption

AI changes the incentives to produce and maintain knowledge. [A recent statement](#) initially signed by 25 Fields Medalists makes this concern vivid in mathematics. When people can obtain answers without undertaking the learning or investigation previously required, they may have less reason to perform activities that benefit others. This raises a question about epistemic institution design in the era of AI: how to best modify existing incentive structures to sustain productive collective inquiry in response of the disruptions of AI?

This question is a natural continuation of my existing work on epistemic institutions. A good answer must take both individual incentives, social epistemic institutions and empirical facts of AI into consideration. My unique background in philosophy, economics and AI makes me particularly suitable to contribute.

4. AI and Social Epistemology

AI creates new settings in which to investigate the epistemic concepts used to understand individual reasoning and collective inquiry. I ask both how philosophical accounts can inform the design of artificial systems and how those systems can help assess the accounts themselves. The following projects develop this reciprocal relationship between AI and social epistemology.

4.1. Agential Uncertainty and Artificial Social Intelligence

Agential uncertainty refers to the uncertainty regarding other agent's behaviors. It's (arguably) distinct from statistical uncertainty, which diminishes as more observations are made. Agential uncertainty is best handled by understanding other agents' intentions, motives, incentives, etc. AI technology has achieved astonishing success with respect to sample uncertainty, but much less advances have been made with respect to agential uncertainty. The purpose of the project is two-folded.

Firstly, I hope to apply insights on agency from philosophy to improve AI's social capacity. In collaboration with AI researchers³, we proposed a novel framework for human decision support based on two philosophical ideas, (i) counterfactual explanation and (ii) computational theory of mind. The resulting system can provide human understandable explanations and intervene based on projected human intention. Secondly, the more theoretical goal is to understand the fundamental difference between agent and environment from an epistemic perspective. How is my uncertainty about others' behaviors

³ "Personalized Decision Supports based on Theory of Mind Modeling and Explainable Reinforcement Learning", 2023 IEEE International Conference on Systems, Man, and Cybernetics (SMC).

different from my uncertainty about tomorrow's weather? This question has been debated in game theory. I hope to contribute in light of advances of AI.

4.2. Propositional Coherence and LLMs

Wen et. al. (2026) proposed Internal Coherence Maximization (ICM) as a method to fine-tune base language models using data generated by base models themselves, to replace the standard Reinforcement Learning from Human Feedback (RLHF). ICM improves the performance of LLMs by increasing the internal coherence ⁴ of LLMs' propositional judgments. This connects directly to existing literature on probabilistic theory of coherence. Existing philosophical literature already relies heavily on computational simulation to test different measures of coherence. ⁵ The new development of LLMs allows us to use LLMs as artificial epistemic agents to test different definitions of coherence. Besides, we can replace the overly simplistic definition of coherence used by Wen et. al. to improve the performance of ICM. It also relates to the more general philosophical question of identifying propositional judgments of LLMs, which in turn relates to the problem of propositional interpretability (i.e., understanding propositional attitudes of LLMs if any) and alignment.

5. Trans-Framework Rationality

I have a long-term research interest in rational and pragmatic comparisons of competing conceptual frameworks, including scientific paradigms in the context of science and (comprehensive) social forms/ideals in the context of ethics. Different conceptual frameworks organize relevant human activities (e.g., scientific research or social interactions) differently with different internal standards of evaluation. Kuhn famously emphasized the incommensurability among scientific paradigms. The same goes for social forms/ideals. ⁶ It's easy to criticize one social ideal based on another, or to compare two based on a third. Yet, to what extent is such comparison rational?

I'm starting this project with a focus on two authors: Michael Friedman and Rahel Jaeggi. Michael Friedman's late work focuses on trans-paradigm rationality, with elaborated historical research into the paradigm change from Newtonian Gravity to General Relativity. Rahel Jaeggi, on the other hand, tries to compare and criticize different forms of life in her *Critique of Forms of Life*. I hope to draw from their work while supplying insights from social epistemology, especially from the pragmatist tradition of Peirce and Bernard Williams among others.

⁴ Here, 'coherence' is understood as how well different propositions hang together, e.g., providing mutual supports or substantial overlap, but not as satisfies the axioms of probability.

⁵ See e.g., Douven and Meijs (2007) and Hartmann and Trpin (2026).

⁶ Think, for a stylized and extreme example, the medieval Mongol elites who think that the highest individual and social good is conquest.